

Nikhil Prabhu

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EDUCATION

University of California, San Diego (UCSD)
B.S. in Computer Science | Cumulative GPA: 3.67

San Diego, CA
Expected Graduation: June 2028

SKILLS

Languages: Python, C++, TypeScript, JavaScript

Core Robotics: SLAM, visual odometry, localization/mapping, path planning, sim-to-real, GPS-denied navigation

Perception/ML: PyTorch, YOLOX, YOLOv8, OpenCV, image segmentation, PPO/MAPPO, behavioral cloning

Simulation + Hardware: MuJoCo, Gymnasium, ROS 2, Three.js, NVIDIA Jetson, embedded systems, Docker, field iteration

EXPERIENCE

Undergraduate Research Assistant — AI for Circuit Invention, UCSD

San Diego, CA

Research Assistant (with Prof. Prasad Gudem & Samvid AI)

March 2026 – Present

- Built an LLM + Cadence simulation closed loop generating and evaluating candidate designs against SOTA benchmarks, optimizing across accuracy and speed.
- Integrated Spectre simulation APIs and explored RL strategies for efficient multimodal design-space search over generated topologies.

Lead Researcher — Foot Drop Ankle-Foot Orthosis (Georgia Tech mmWave Lab)

Atlanta, GA

IEEE ICHCI 2024 Presenter & Published Author

May 2024 – Nov 2024

- Built real-time EMG sensing + actuation on Arduino embedded hardware (R = 0.968 correlation, 23% gait precision gain); first-author paper at IEEE ICHCI 2024.

Lead Researcher — Brain-Computer Interface (Stanford Lee-Messer Lab)

Stanford, CA

Stanford University School of Medicine

Oct 2023 – Dec 2024

- Benchmarked P300 EEG models converting noisy neural signals into reliable assistive-control output; piloted with 12 users and shipped a React Native frontend for motor-impaired students.

ROBOTICS PROJECTS

DroneOS

1st Overall, Bow Capital × DS3 × SIC Hackathon (UCSD)

2026

- Built a GPS-denied autonomous platform with SLAM, visual odometry, geo-referenced 3D Gaussian Splatting maps, and on-device friend-or-foe classification over YOLOX + Norfair perception.
- Trained swarm policies (MAPPO + behavioral cloning) for coordination without GPS or comms that could be jammed; validated sim-to-real transfer from simulation to flight.

SODIUM — Autonomous Care Robot

1st Overall, DiamondHacks 2026

April 2026

- Built an autonomous ROS 2 stack on Jetson-class embedded compute with wake-word detection, on-device perception/control, medication reminders, and crisis detection plus a real-time caregiver dashboard (Svelte 5 / Bun).

Battle Angel — Sim-to-Real Policy Training

Google Gemini 3.5 Hackathon

2026

- Trained vectorized PPO policies for a Unitree G1 humanoid across eight MuJoCo disaster environments with Gemini-generated scene layouts and live rollout GIF replay.

NIGEL — SLAM Mission-Control for First Responders

1st Overall, Transpose / Compiled-5 UCSD Hackathon

2026

- Built a dispatcher dashboard ingesting Odin helmet VSLAM point clouds, multi-unit video feeds, and tactical mapping over WebSockets (Next.js, Three.js, ROS 2, C++).

PUBLICATIONS

- **Real-Time EMG Control for Assistive Devices** — IEEE ICHCI 2024.
- **Co-Heritability of Antibiotic Resistance in TB Evolution** — Bioinformatics & Biomedical Technology, 2025.